Newman-Penrose Formalism Calculations

Stephani et al., Equation (31.41_1)

Metric

$$ds^2 = -H^2 \mathrm{d}u \otimes \mathrm{d}u + \left(-\frac{l^2 - z^2}{l^2 + z^2}\right) \mathrm{d}u \otimes \mathrm{d}v + \left(-\frac{2\,mvz^2}{l^4 + 2\,l^2z^2 + z^4}\right) \mathrm{d}u \otimes \mathrm{d}z + \left(\frac{2\,lmvz}{l^4 + 2\,l^2z^2 + z^4}\right) \mathrm{d}u \otimes \mathrm{d}y + \left(-\frac{l^2 - z^2}{l^2 + z^2}\right) \mathrm{d}v \otimes \mathrm{d}u + \left(-\frac{2\,mvz^2}{l^4 + 2\,l^2z^2 + z^4}\right) \mathrm{d}z \otimes \mathrm{d}u + \mathrm{d}z \otimes \mathrm{d}z + \left(\frac{2\,lmvz}{l^4 + 2\,l^2z^2 + z^4}\right) \mathrm{d}y \otimes \mathrm{d}u + \left(\frac{l^2 + z^2}{2\,mz}\right) \mathrm{d}y \otimes \mathrm{d}y$$

Coordinates

$(x^\mu) = (u, v, z, y)$

Metric Functions

$$H = -\frac{4\,l^2mv^2z}{l^6 + 3\,l^4z^2 + 3\,l^2z^4 + z^6}$$

Null Tetrad

Covariant components

$l = (1)\ du$

$$n = \left(\frac{1}{2}\,H^2\right)\ du + \left(\frac{l^2}{l^2 + z^2} - \frac{z^2}{l^2 + z^2}\right)\ dv + \left(\frac{2\,mvz^2}{l^4 + 2\,l^2z^2 + z^4}\right)\ dz + \left(-\frac{2\,lmvz}{l^4 + 2\,l^2z^2 + z^4}\right)\ dy$$

$$m = \left(\sqrt{\frac{1}{2}}\right)\ dz + \left(\frac{i\,\sqrt{2}\sqrt{\frac{1}{2}}\sqrt{l^2 + z^2}}{2\,\sqrt{m}\sqrt{z}}\right)\ dy$$

$$\overline{m} = \left(\sqrt{\frac{1}{2}}\right) dz + \left(-\frac{i\sqrt{2}\sqrt{\frac{1}{2}}\sqrt{l^2+z^2}}{2\sqrt{m}\sqrt{z}}\right) dy$$

Contravariant components

$$l = \left(-\frac{l^2}{l^2-z^2} - \frac{z^2}{l^2-z^2}\right) dv$$

$$n = (-1)\, du + \left(\frac{l^2H^2}{2\left(l^2-z^2\right)} + \frac{z^2H^2}{2\left(l^2-z^2\right)}\right) dv$$

$$m = \left(-\frac{\sqrt{2}l^2mvz^2}{l^6+l^4z^2-l^2z^4-z^6} - \frac{\sqrt{2}mvz^4}{l^6+l^4z^2-l^2z^4-z^6} + \frac{2i\sqrt{l^2+z^2}lm^{\frac{3}{2}}vz^{\frac{3}{2}}}{l^6+l^4z^2-l^2z^4-z^6}\right) dv + \left(\sqrt{\frac{1}{2}}\right) dz + \left(\frac{i\sqrt{m}\sqrt{z}}{\sqrt{l^2+z^2}}\right) dy$$

$$\overline{m} = \left(-\frac{\sqrt{2}l^2mvz^2}{l^6+l^4z^2-l^2z^4-z^6} - \frac{\sqrt{2}mvz^4}{l^6+l^4z^2-l^2z^4-z^6} - \frac{2i\sqrt{l^2+z^2}lm^{\frac{3}{2}}vz^{\frac{3}{2}}}{l^6+l^4z^2-l^2z^4-z^6}\right) dv + \left(\sqrt{\frac{1}{2}}\right) dz + \left(-\frac{i\sqrt{m}\sqrt{z}}{\sqrt{l^2+z^2}}\right) dy$$

Directional Derivatives

$$D = -\frac{l^2}{l^2-z^2} - \frac{z^2}{l^2-z^2}\,\partial_v$$

$$\Delta = -\partial_u + \frac{l^2H^2}{2\left(l^2-z^2\right)} + \frac{z^2H^2}{2\left(l^2-z^2\right)}\,\partial_v$$

$$\delta = -\frac{\sqrt{2}l^2mvz^2}{l^6+l^4z^2-l^2z^4-z^6} - \frac{\sqrt{2}mvz^4}{l^6+l^4z^2-l^2z^4-z^6} + \frac{2i\sqrt{l^2+z^2}lm^{\frac{3}{2}}vz^{\frac{3}{2}}}{l^6+l^4z^2-l^2z^4-z^6}\,\partial_v + \sqrt{\frac{1}{2}}\,\partial_z + \frac{i\sqrt{m}\sqrt{z}}{\sqrt{l^2+z^2}}\,\partial_y$$

$$\overline{\delta} = -\frac{\sqrt{2}l^2mvz^2}{l^6+l^4z^2-l^2z^4-z^6} - \frac{\sqrt{2}mvz^4}{l^6+l^4z^2-l^2z^4-z^6} - \frac{2i\sqrt{l^2+z^2}lm^{\frac{3}{2}}vz^{\frac{3}{2}}}{l^6+l^4z^2-l^2z^4-z^6}\,\partial_v + \sqrt{\frac{1}{2}}\,\partial_z - \frac{i\sqrt{m}\sqrt{z}}{\sqrt{l^2+z^2}}\,\partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{i\sqrt{2}\sqrt{l^2+z^2}l^5m^{\frac{3}{2}}v\sqrt{z}}{l^{10}+3\,l^8z^2+2\,l^6z^4-2\,l^4z^6-3\,l^2z^8-z^{10}} + \frac{3i\sqrt{2}\sqrt{l^2+z^2}lm^{\frac{3}{2}}vz^{\frac{9}{2}}}{l^{10}+3\,l^8z^2+2\,l^6z^4-2\,l^4z^6-3\,l^2z^8-z^{10}}$$

$$\lambda = 0$$

$$\nu = \frac{\sqrt{2}l^4H\frac{\partial}{\partial z}H}{2\,(l^4-z^4)} - \frac{\sqrt{2}z^4H\frac{\partial}{\partial z}H}{2\,(l^4-z^4)} - \frac{2i\,lm^{\frac{3}{2}}vz^{\frac{3}{2}}H\frac{\partial}{\partial v}H}{(l^4-z^4)\sqrt{l^2+z^2}} - \frac{\sqrt{2}mvz^2H\frac{\partial}{\partial v}H}{l^4-z^4} + \frac{\sqrt{2}l^2zH^2}{l^4-z^4} + \frac{i\,lm^{\frac{3}{2}}z^{\frac{3}{2}}H^2}{(l^4-z^4)\sqrt{l^2+z^2}} + \frac{\sqrt{2}mz^2H^2}{2\,(l^4-z^4)}$$

$$\alpha = \frac{\sqrt{2}l^6}{8\,(l^6z+l^4z^3-l^2z^5-z^7)} + \frac{3\sqrt{2}l^4z^2}{8\,(l^6z+l^4z^3-l^2z^5-z^7)} + \frac{\sqrt{2}l^2mz^3}{4\,(l^6z+l^4z^3-l^2z^5-z^7)} + \frac{3\sqrt{2}l^2z^4}{8\,(l^6z+l^4z^3-l^2z^5-z^7)} + \frac{\sqrt{2}mz^5}{4\,(l^6z+l^4z^3-l^2z^5-z^7)} + \frac{\sqrt{2}z^6}{8\,(l^6z+l^4z^3-l^2z^5-z^7)} + \frac{i\sqrt{l^2+z^2}lm^{\frac{3}{2}}z^{\frac{5}{2}}}{2\,(l^6z+l^4z^3-l^2z^5-z^7)}$$

$$\beta = -\frac{\sqrt{2}l^4}{8\,(l^4z-z^5)} + \frac{3\sqrt{2}l^2z^2}{4\,(l^4z-z^5)} - \frac{i\,lm^{\frac{3}{2}}z^{\frac{5}{2}}}{2\,(l^4z-z^5)\sqrt{l^2+z^2}} + \frac{\sqrt{2}mz^3}{4\,(l^4z-z^5)} - \frac{\sqrt{2}z^4}{8\,(l^4z-z^5)}$$

$$\tau = -\frac{i\sqrt{2}lm^{\frac{3}{2}}z^{\frac{3}{2}}}{(\sqrt{2}l^4-\sqrt{2}z^4)\sqrt{l^2+z^2}} + \frac{2\,l^2z}{\sqrt{2}l^4-\sqrt{2}z^4} + \frac{mz^2}{\sqrt{2}l^4-\sqrt{2}z^4}$$

$$\pi = -\frac{2\,l^4z}{\sqrt{2}l^6+\sqrt{2}l^4z^2-\sqrt{2}l^2z^4-\sqrt{2}z^6} - \frac{i\sqrt{2}\sqrt{l^2+z^2}lm^{\frac{3}{2}}z^{\frac{3}{2}}}{\sqrt{2}l^6+\sqrt{2}l^4z^2-\sqrt{2}l^2z^4-\sqrt{2}z^6} - \frac{l^2mz^2}{\sqrt{2}l^6+\sqrt{2}l^4z^2-\sqrt{2}l^2z^4-\sqrt{2}z^6} - \frac{2\,l^2z^3}{\sqrt{2}l^6+\sqrt{2}l^4z^2-\sqrt{2}l^2z^4-\sqrt{2}z^6} - \frac{mz^4}{\sqrt{2}l^6+\sqrt{2}l^4z^2-\sqrt{2}l^2z^4-\sqrt{2}z^6}$$

$\epsilon = 0$

$$\gamma = -\frac{\sqrt{2}l^8 H \frac{\partial}{\partial v} H}{2\left(\sqrt{2}l^8 + 2\sqrt{2}l^6 z^2 - 2\sqrt{2}l^2 z^6 - \sqrt{2}z^8\right)} - \frac{2\sqrt{2}l^6 z^2 H \frac{\partial}{\partial v} H}{\sqrt{2}l^8 + 2\sqrt{2}l^6 z^2 - 2\sqrt{2}l^2 z^6 - \sqrt{2}z^8} - \frac{3\sqrt{2}l^4 z^4 H \frac{\partial}{\partial v} H}{\sqrt{2}l^8 + 2\sqrt{2}l^6 z^2 - 2\sqrt{2}l^2 z^6 - \sqrt{2}z^8} - \frac{2\sqrt{2}l^2 z^6 H \frac{\partial}{\partial v} H}{\sqrt{2}l^8 + 2\sqrt{2}l^6 z^2 - 2\sqrt{2}l^2 z^6 - \sqrt{2}z^8}$$
$$- \frac{\sqrt{2}z^8 H \frac{\partial}{\partial v} H}{2\left(\sqrt{2}l^8 + 2\sqrt{2}l^6 z^2 - 2\sqrt{2}l^2 z^6 - \sqrt{2}z^8\right)} + \frac{i\, l^5 m^{\frac{3}{2}} v \sqrt{z}}{\left(\sqrt{2}l^8 + 2\sqrt{2}l^6 z^2 - 2\sqrt{2}l^2 z^6 - \sqrt{2}z^8\right)\sqrt{l^2 + z^2}} + \frac{3i\, l m^{\frac{3}{2}} v z^{\frac{9}{2}}}{\left(\sqrt{2}l^8 + 2\sqrt{2}l^6 z^2 - 2\sqrt{2}l^2 z^6 - \sqrt{2}z^8\right)\sqrt{l^2 + z^2}}$$

Ricci Tensor Components

$\Phi_{00} = 0$

$\Phi_{01} = 0$

$$\Phi_{02} = \frac{3\, l^8}{2\left(l^{10} + l^8 z^2 - 2\, l^6 z^4 - 2\, l^4 z^6 + l^2 z^8 + z^{10}\right)} - \frac{i\sqrt{2}\sqrt{l^2 + z^2} l^5 m^{\frac{3}{2}} \sqrt{z}}{l^{10} + l^8 z^2 - 2\, l^6 z^4 - 2\, l^4 z^6 + l^2 z^8 + z^{10}} + \frac{5\, l^6 m z}{4\left(l^{10} + l^8 z^2 - 2\, l^6 z^4 - 2\, l^4 z^6 + l^2 z^8 + z^{10}\right)}$$
$$- \frac{3\, l^6 z^2}{2\left(l^{10} + l^8 z^2 - 2\, l^6 z^4 - 2\, l^4 z^6 + l^2 z^8 + z^{10}\right)} + \frac{3i\sqrt{2}\sqrt{l^2 + z^2} l^3 m^{\frac{3}{2}} z^{\frac{5}{2}}}{l^{10} + l^8 z^2 - 2\, l^6 z^4 - 2\, l^4 z^6 + l^2 z^8 + z^{10}} - \frac{5\, l^4 m z^3}{4\left(l^{10} + l^8 z^2 - 2\, l^6 z^4 - 2\, l^4 z^6 + l^2 z^8 + z^{10}\right)}$$
$$+ \frac{l^2 m^3 z^3}{l^{10} + l^8 z^2 - 2\, l^6 z^4 - 2\, l^4 z^6 + l^2 z^8 + z^{10}} + \frac{i\sqrt{2}\sqrt{l^2 + z^2} l m^{\frac{5}{2}} z^{\frac{7}{2}}}{l^{10} + l^8 z^2 - 2\, l^6 z^4 - 2\, l^4 z^6 + l^2 z^8 + z^{10}} + \frac{l^4 z^4}{2\left(l^{10} + l^8 z^2 - 2\, l^6 z^4 - 2\, l^4 z^6 + l^2 z^8 + z^{10}\right)}$$
$$- \frac{l^2 m^2 z^4}{2\left(l^{10} + l^8 z^2 - 2\, l^6 z^4 - 2\, l^4 z^6 + l^2 z^8 + z^{10}\right)} - \frac{2i\sqrt{2}\sqrt{l^2 + z^2} l m^{\frac{3}{2}} z^{\frac{9}{2}}}{l^{10} + l^8 z^2 - 2\, l^6 z^4 - 2\, l^4 z^6 + l^2 z^8 + z^{10}} - \frac{5\, l^2 m z^5}{4\left(l^{10} + l^8 z^2 - 2\, l^6 z^4 - 2\, l^4 z^6 + l^2 z^8 + z^{10}\right)}$$
$$+ \frac{7\, l^2 z^6}{2\left(l^{10} + l^8 z^2 - 2\, l^6 z^4 - 2\, l^4 z^6 + l^2 z^8 + z^{10}\right)} - \frac{m^2 z^6}{2\left(l^{10} + l^8 z^2 - 2\, l^6 z^4 - 2\, l^4 z^6 + l^2 z^8 + z^{10}\right)} + \frac{5\, m z^7}{4\left(l^{10} + l^8 z^2 - 2\, l^6 z^4 - 2\, l^4 z^6 + l^2 z^8 + z^{10}\right)}$$

$\Phi_{10} = 0$

$$
\begin{aligned}
\Phi_{11} = {} & \frac{l^{10} z^2 \frac{\partial}{\partial v} H^2}{4\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{5\,l^8 z^4 \frac{\partial}{\partial v} H^2}{4\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{5\,l^6 z^6 \frac{\partial}{\partial v} H^2}{2\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} \\
& + \frac{5\,l^4 z^8 \frac{\partial}{\partial v} H^2}{2\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{5\,l^2 z^{10} \frac{\partial}{\partial v} H^2}{4\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{z^{12} \frac{\partial}{\partial v} H^2}{4\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} \\
& + \frac{l^{10} z^2 H \frac{\partial^2}{(\partial v)^2} H}{4\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{5\,l^8 z^4 H \frac{\partial^2}{(\partial v)^2} H}{4\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{5\,l^6 z^6 H \frac{\partial^2}{(\partial v)^2} H}{2\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} \\
& + \frac{5\,l^4 z^8 H \frac{\partial^2}{(\partial v)^2} H}{2\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{5\,l^2 z^{10} H \frac{\partial^2}{(\partial v)^2} H}{4\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{z^{12} H \frac{\partial^2}{(\partial v)^2} H}{4\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} \\
& - \frac{3\,l^{10}}{16\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} - \frac{3\,l^8 z^2}{16\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} \\
& + \frac{13\,l^6 z^4}{8\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{l^4 m z^5}{l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12}} + \frac{l^2 m^3 z^5}{2\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} \\
& + \frac{9\,l^4 z^6}{8\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{l^2 m^2 z^6}{4\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{l^2 m z^7}{l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12}} \\
& - \frac{7\,l^2 z^8}{16\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{m^2 z^8}{4\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{z^{10}}{16\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})}
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{12} = &-\frac{i\sqrt{2}l^{11}m^{\frac{3}{2}}vz^{\frac{3}{2}}\frac{\partial}{\partial v}H^2}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
&-\frac{5i\sqrt{2}l^{9}m^{\frac{3}{2}}vz^{\frac{7}{2}}\frac{\partial}{\partial v}H^2}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
&-\frac{10i\sqrt{2}l^{7}m^{\frac{3}{2}}vz^{\frac{11}{2}}\frac{\partial}{\partial v}H^2}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
&-\frac{10i\sqrt{2}l^{5}m^{\frac{3}{2}}vz^{\frac{15}{2}}\frac{\partial}{\partial v}H^2}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
&-\frac{5i\sqrt{2}l^{3}m^{\frac{3}{2}}vz^{\frac{19}{2}}\frac{\partial}{\partial v}H^2}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
&-\frac{i\sqrt{2}lm^{\frac{3}{2}}vz^{\frac{23}{2}}\frac{\partial}{\partial v}H^2}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
&-\frac{i\sqrt{2}l^{11}m^{\frac{3}{2}}vz^{\frac{3}{2}}H\frac{\partial^2}{(\partial v)^2}H}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
&-\frac{5i\sqrt{2}l^{9}m^{\frac{3}{2}}vz^{\frac{7}{2}}H\frac{\partial^2}{(\partial v)^2}H}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
&-\frac{10i\sqrt{2}l^{7}m^{\frac{3}{2}}vz^{\frac{11}{2}}H\frac{\partial^2}{(\partial v)^2}H}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
&-\frac{10i\sqrt{2}l^{5}m^{\frac{3}{2}}vz^{\frac{15}{2}}H\frac{\partial^2}{(\partial v)^2}H}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
&-\frac{5i\sqrt{2}l^{3}m^{\frac{3}{2}}vz^{\frac{19}{2}}H\frac{\partial^2}{(\partial v)^2}H}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
&-\frac{i\sqrt{2}lm^{\frac{3}{2}}vz^{\frac{23}{2}}H\frac{\partial^2}{(\partial v)^2}H}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
&-\frac{l^{14}H\frac{\partial^2}{\partial z\partial v}H}{2\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)} \\
&-\frac{5\,l^{12}z^2H\frac{\partial^2}{\partial z\partial v}H}{2\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)} \\
&-\frac{9\,l^{10}z^4H\frac{\partial^2}{\partial z\partial v}H}{2\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)} - \cdots
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{20} = {} & \frac{3\,l^8}{2\,(l^{10}+l^8z^2-2\,l^6z^4-2\,l^4z^6+l^2z^8+z^{10})} + \frac{i\,\sqrt{2}\sqrt{l^2+z^2}l^5m^{\frac{3}{2}}\sqrt{z}}{l^{10}+l^8z^2-2\,l^6z^4-2\,l^4z^6+l^2z^8+z^{10}} + \frac{5\,l^6mz}{4\,(l^{10}+l^8z^2-2\,l^6z^4-2\,l^4z^6+l^2z^8+z^{10})} \\
& - \frac{3\,l^6z^2}{2\,(l^{10}+l^8z^2-2\,l^6z^4-2\,l^4z^6+l^2z^8+z^{10})} - \frac{3i\,\sqrt{2}\sqrt{l^2+z^2}l^3m^{\frac{3}{2}}z^{\frac{5}{2}}}{l^{10}+l^8z^2-2\,l^6z^4-2\,l^4z^6+l^2z^8+z^{10}} - \frac{5\,l^4mz^3}{4\,(l^{10}+l^8z^2-2\,l^6z^4-2\,l^4z^6+l^2z^8+z^{10})} \\
& + \frac{l^2m^3z^3}{l^{10}+l^8z^2-2\,l^6z^4-2\,l^4z^6+l^2z^8+z^{10}} - \frac{i\,\sqrt{2}\sqrt{l^2+z^2}lm^{\frac{5}{2}}z^{\frac{7}{2}}}{l^{10}+l^8z^2-2\,l^6z^4-2\,l^4z^6+l^2z^8+z^{10}} + \frac{l^4z^4}{2\,(l^{10}+l^8z^2-2\,l^6z^4-2\,l^4z^6+l^2z^8+z^{10})} \\
& - \frac{l^2m^2z^4}{2\,(l^{10}+l^8z^2-2\,l^6z^4-2\,l^4z^6+l^2z^8+z^{10})} + \frac{2i\,\sqrt{2}\sqrt{l^2+z^2}lm^{\frac{3}{2}}z^{\frac{9}{2}}}{l^{10}+l^8z^2-2\,l^6z^4-2\,l^4z^6+l^2z^8+z^{10}} - \frac{5\,l^2mz^5}{4\,(l^{10}+l^8z^2-2\,l^6z^4-2\,l^4z^6+l^2z^8+z^{10})} \\
& + \frac{7\,l^2z^6}{2\,(l^{10}+l^8z^2-2\,l^6z^4-2\,l^4z^6+l^2z^8+z^{10})} - \frac{m^2z^6}{2\,(l^{10}+l^8z^2-2\,l^6z^4-2\,l^4z^6+l^2z^8+z^{10})} + \frac{5\,mz^7}{4\,(l^{10}+l^8z^2-2\,l^6z^4-2\,l^4z^6+l^2z^8+z^{10})}
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{21} = {} & \frac{i\sqrt{2}l^{11}m^{\frac{3}{2}}vz^{\frac{3}{2}}\frac{\partial}{\partial v}H^2}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
& + \frac{5i\sqrt{2}l^{9}m^{\frac{3}{2}}vz^{\frac{7}{2}}\frac{\partial}{\partial v}H^2}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
& + \frac{10i\sqrt{2}l^{7}m^{\frac{3}{2}}vz^{\frac{11}{2}}\frac{\partial}{\partial v}H^2}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
& + \frac{10i\sqrt{2}l^{5}m^{\frac{3}{2}}vz^{\frac{15}{2}}\frac{\partial}{\partial v}H^2}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
& + \frac{5i\sqrt{2}l^{3}m^{\frac{3}{2}}vz^{\frac{19}{2}}\frac{\partial}{\partial v}H^2}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
& + \frac{i\sqrt{2}lm^{\frac{3}{2}}vz^{\frac{23}{2}}\frac{\partial}{\partial v}H^2}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
& + \frac{i\sqrt{2}l^{11}m^{\frac{3}{2}}vz^{\frac{3}{2}}H\frac{\partial^2}{(\partial v)^2}H}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
& + \frac{5i\sqrt{2}l^{9}m^{\frac{3}{2}}vz^{\frac{7}{2}}H\frac{\partial^2}{(\partial v)^2}H}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
& + \frac{10i\sqrt{2}l^{7}m^{\frac{3}{2}}vz^{\frac{11}{2}}H\frac{\partial^2}{(\partial v)^2}H}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
& + \frac{10i\sqrt{2}l^{5}m^{\frac{3}{2}}vz^{\frac{15}{2}}H\frac{\partial^2}{(\partial v)^2}H}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
& + \frac{5i\sqrt{2}l^{3}m^{\frac{3}{2}}vz^{\frac{19}{2}}H\frac{\partial^2}{(\partial v)^2}H}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
& + \frac{i\sqrt{2}lm^{\frac{3}{2}}vz^{\frac{23}{2}}H\frac{\partial^2}{(\partial v)^2}H}{\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)\sqrt{l^2+z^2}} \\
& - \frac{l^{14}H\frac{\partial^2}{\partial z\partial v}H}{2\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)} \\
& - \frac{5\,l^{12}z^2H\frac{\partial^2}{\partial z\partial v}H}{2\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)} \\
& - \frac{9\,l^{10}z^4H\frac{\partial^2}{\partial z\partial v}H}{2\left(\sqrt{2}l^{14}+3\sqrt{2}l^{12}z^2+\sqrt{2}l^{10}z^4-5\sqrt{2}l^8z^6-5\sqrt{2}l^6z^8+\sqrt{2}l^4z^{10}+3\sqrt{2}l^2z^{12}+\sqrt{2}z^{14}\right)} - \cdots
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{22} = {} & \frac{l^{18} z \frac{\partial}{\partial z} H^2}{2\left(l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}\right)} + \frac{5\, l^{16} z^3 \frac{\partial}{\partial z} H^2}{2\left(l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}\right)} \\
& + \frac{4\, l^{14} z^5 \frac{\partial}{\partial z} H^2}{l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}} - \frac{7\, l^{10} z^9 \frac{\partial}{\partial z} H^2}{l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}} \\
& - \frac{7\, l^8 z^{11} \frac{\partial}{\partial z} H^2}{l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}} + \frac{4\, l^4 z^{15} \frac{\partial}{\partial z} H^2}{l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}} \\
& + \frac{5\, l^2 z^{17} \frac{\partial}{\partial z} H^2}{2\left(l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}\right)} + \frac{z^{19} \frac{\partial}{\partial z} H^2}{2\left(l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}\right)} \\
& + \frac{l^{18} z H \frac{\partial^2}{(\partial z)^2} H}{2\left(l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}\right)} + \frac{5\, l^{16} z^3 H \frac{\partial^2}{(\partial z)^2} H}{2\left(l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}\right)} \\
& + \frac{4\, l^{14} z^5 H \frac{\partial^2}{(\partial z)^2} H}{l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}} - \frac{7\, l^{10} z^9 H \frac{\partial^2}{(\partial z)^2} H}{l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}} \\
& - \frac{7\, l^8 z^{11} H \frac{\partial^2}{(\partial z)^2} H}{l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}} + \frac{4\, l^4 z^{15} H \frac{\partial^2}{(\partial z)^2} H}{l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}} \\
& + \frac{5\, l^2 z^{17} H \frac{\partial^2}{(\partial z)^2} H}{2\left(l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}\right)} + \frac{z^{19} H \frac{\partial^2}{(\partial z)^2} H}{2\left(l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}\right)} \\
& - \frac{2\, l^{14} m v z^3 H \frac{\partial^2}{\partial z \partial v} H}{l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}} - \frac{10\, l^{12} m v z^5 H \frac{\partial^2}{\partial z \partial v} H}{l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}} \\
& - \frac{18\, l^{10} m v z^7 H \frac{\partial^2}{\partial z \partial v} H}{l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}} - \frac{10\, l^8 m v z^9 H \frac{\partial^2}{\partial z \partial v} H}{l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}} \\
& + \frac{10\, l^6 m v z^{11} H \frac{\partial^2}{\partial z \partial v} H}{l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}} + \frac{18\, l^4 m v z^{13} H \frac{\partial^2}{\partial z \partial v} H}{l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}} \\
& + \frac{10\, l^2 m v z^{15} H \frac{\partial^2}{\partial z \partial v} H}{l^{18} z + 5\, l^{16} z^3 + 8\, l^{14} z^5 - 14\, l^{10} z^9 - 14\, l^8 z^{11} + 8\, l^4 z^{15} + 5\, l^2 z^{17} + z^{19}} + \cdots
\end{aligned}
$$

$$
\begin{aligned}
\Lambda = & -\frac{l^{10}z^2\frac{\partial}{\partial v}H^2}{12\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} - \frac{5\,l^8z^4\frac{\partial}{\partial v}H^2}{12\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} - \frac{5\,l^6z^6\frac{\partial}{\partial v}H^2}{6\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} \\
& - \frac{5\,l^4z^8\frac{\partial}{\partial v}H^2}{6\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} - \frac{5\,l^2z^{10}\frac{\partial}{\partial v}H^2}{12\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} - \frac{z^{12}\frac{\partial}{\partial v}H^2}{12\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} \\
& - \frac{l^{10}z^2H\frac{\partial^2}{(\partial v)^2}H}{12\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} - \frac{5\,l^8z^4H\frac{\partial^2}{(\partial v)^2}H}{12\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} - \frac{5\,l^6z^6H\frac{\partial^2}{(\partial v)^2}H}{6\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} \\
& - \frac{5\,l^4z^8H\frac{\partial^2}{(\partial v)^2}H}{6\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} - \frac{5\,l^2z^{10}H\frac{\partial^2}{(\partial v)^2}H}{12\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} - \frac{z^{12}H\frac{\partial^2}{(\partial v)^2}H}{12\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} \\
& - \frac{l^{10}}{16\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} + \frac{5\,l^8z^2}{48\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} \\
& + \frac{l^6mz^3}{4\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} - \frac{7\,l^6z^4}{24\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} - \frac{7\,l^4mz^5}{12\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} \\
& - \frac{l^2m^3z^5}{2\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} + \frac{5\,l^4z^6}{24\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} - \frac{l^2m^2z^6}{4\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} \\
& - \frac{7\,l^2mz^7}{12\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} + \frac{11\,l^2z^8}{16\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} \\
& - \frac{m^2z^8}{4\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} + \frac{mz^9}{4\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)} + \frac{z^{10}}{48\left(l^{10}z^2+l^8z^4-2\,l^6z^6-2\,l^4z^8+l^2z^{10}+z^{12}\right)}
\end{aligned}
$$

Weyl Tensor Components

$\Psi_0 = 0$

$\Psi_1 = 0$

$$
\begin{aligned}
\Psi_2 = {} & \frac{l^{10} z^2 \frac{\partial}{\partial v} H^2}{6\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{5\,l^8 z^4 \frac{\partial}{\partial v} H^2}{6\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{5\,l^6 z^6 \frac{\partial}{\partial v} H^2}{3\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} \\
& + \frac{5\,l^4 z^8 \frac{\partial}{\partial v} H^2}{3\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{5\,l^2 z^{10} \frac{\partial}{\partial v} H^2}{6\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{z^{12} \frac{\partial}{\partial v} H^2}{6\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} \\
& + \frac{l^{10} z^2 H \frac{\partial^2}{(\partial v)^2} H}{6\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{5\,l^8 z^4 H \frac{\partial^2}{(\partial v)^2} H}{6\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{5\,l^6 z^6 H \frac{\partial^2}{(\partial v)^2} H}{3\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} \\
& + \frac{5\,l^4 z^8 H \frac{\partial^2}{(\partial v)^2} H}{3\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{5\,l^2 z^{10} H \frac{\partial^2}{(\partial v)^2} H}{6\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{z^{12} H \frac{\partial^2}{(\partial v)^2} H}{6\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} \\
& + \frac{l^{10}}{8\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{7\,l^8 z^2}{24\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} - \frac{i\,\sqrt{2}\sqrt{l^2 + z^2}\, l^5 m^{\frac{3}{2}} z^{\frac{5}{2}}}{2\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} \\
& + \frac{l^6 m z^3}{4\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{l^6 z^4}{12\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{5\,l^4 m z^5}{12\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} \\
& + \frac{13\,l^4 z^6}{12\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} - \frac{3 i\,\sqrt{2}\sqrt{l^2 + z^2}\, l m^{\frac{3}{2}} z^{\frac{13}{2}}}{2\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{5\,l^2 m z^7}{12\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} \\
& + \frac{9\,l^2 z^8}{8\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} + \frac{m z^9}{4\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})} - \frac{z^{10}}{24\,(l^{10} z^2 + l^8 z^4 - 2\,l^6 z^6 - 2\,l^4 z^8 + l^2 z^{10} + z^{12})}
\end{aligned}
$$

$$
\begin{aligned}
\Psi_3 = &-\frac{l^{14} H \frac{\partial^2}{\partial z \partial v} H}{2\left(\sqrt{2} l^{14} + 3\sqrt{2} l^{12} z^2 + \sqrt{2} l^{10} z^4 - 5\sqrt{2} l^8 z^6 - 5\sqrt{2} l^6 z^8 + \sqrt{2} l^4 z^{10} + 3\sqrt{2} l^2 z^{12} + \sqrt{2} z^{14}\right)} \\
&- \frac{5\, l^{12} z^2 H \frac{\partial^2}{\partial z \partial v} H}{2\left(\sqrt{2} l^{14} + 3\sqrt{2} l^{12} z^2 + \sqrt{2} l^{10} z^4 - 5\sqrt{2} l^8 z^6 - 5\sqrt{2} l^6 z^8 + \sqrt{2} l^4 z^{10} + 3\sqrt{2} l^2 z^{12} + \sqrt{2} z^{14}\right)} \\
&- \frac{9\, l^{10} z^4 H \frac{\partial^2}{\partial z \partial v} H}{2\left(\sqrt{2} l^{14} + 3\sqrt{2} l^{12} z^2 + \sqrt{2} l^{10} z^4 - 5\sqrt{2} l^8 z^6 - 5\sqrt{2} l^6 z^8 + \sqrt{2} l^4 z^{10} + 3\sqrt{2} l^2 z^{12} + \sqrt{2} z^{14}\right)} \\
&- \frac{5\, l^8 z^6 H \frac{\partial^2}{\partial z \partial v} H}{2\left(\sqrt{2} l^{14} + 3\sqrt{2} l^{12} z^2 + \sqrt{2} l^{10} z^4 - 5\sqrt{2} l^8 z^6 - 5\sqrt{2} l^6 z^8 + \sqrt{2} l^4 z^{10} + 3\sqrt{2} l^2 z^{12} + \sqrt{2} z^{14}\right)} \\
&+ \frac{5\, l^6 z^8 H \frac{\partial^2}{\partial z \partial v} H}{2\left(\sqrt{2} l^{14} + 3\sqrt{2} l^{12} z^2 + \sqrt{2} l^{10} z^4 - 5\sqrt{2} l^8 z^6 - 5\sqrt{2} l^6 z^8 + \sqrt{2} l^4 z^{10} + 3\sqrt{2} l^2 z^{12} + \sqrt{2} z^{14}\right)} \\
&+ \frac{9\, l^4 z^{10} H \frac{\partial^2}{\partial z \partial v} H}{2\left(\sqrt{2} l^{14} + 3\sqrt{2} l^{12} z^2 + \sqrt{2} l^{10} z^4 - 5\sqrt{2} l^8 z^6 - 5\sqrt{2} l^6 z^8 + \sqrt{2} l^4 z^{10} + 3\sqrt{2} l^2 z^{12} + \sqrt{2} z^{14}\right)} \\
&+ \frac{5\, l^2 z^{12} H \frac{\partial^2}{\partial z \partial v} H}{2\left(\sqrt{2} l^{14} + 3\sqrt{2} l^{12} z^2 + \sqrt{2} l^{10} z^4 - 5\sqrt{2} l^8 z^6 - 5\sqrt{2} l^6 z^8 + \sqrt{2} l^4 z^{10} + 3\sqrt{2} l^2 z^{12} + \sqrt{2} z^{14}\right)} \\
&+ \frac{z^{14} H \frac{\partial^2}{\partial z \partial v} H}{2\left(\sqrt{2} l^{14} + 3\sqrt{2} l^{12} z^2 + \sqrt{2} l^{10} z^4 - 5\sqrt{2} l^8 z^6 - 5\sqrt{2} l^6 z^8 + \sqrt{2} l^4 z^{10} + 3\sqrt{2} l^2 z^{12} + \sqrt{2} z^{14}\right)} \\
&- \frac{l^{14} \frac{\partial}{\partial z} H \frac{\partial}{\partial v} H}{2\left(\sqrt{2} l^{14} + 3\sqrt{2} l^{12} z^2 + \sqrt{2} l^{10} z^4 - 5\sqrt{2} l^8 z^6 - 5\sqrt{2} l^6 z^8 + \sqrt{2} l^4 z^{10} + 3\sqrt{2} l^2 z^{12} + \sqrt{2} z^{14}\right)} \\
&- \frac{5\, l^{12} z^2 \frac{\partial}{\partial z} H \frac{\partial}{\partial v} H}{2\left(\sqrt{2} l^{14} + 3\sqrt{2} l^{12} z^2 + \sqrt{2} l^{10} z^4 - 5\sqrt{2} l^8 z^6 - 5\sqrt{2} l^6 z^8 + \sqrt{2} l^4 z^{10} + 3\sqrt{2} l^2 z^{12} + \sqrt{2} z^{14}\right)} \\
&- \frac{9\, l^{10} z^4 \frac{\partial}{\partial z} H \frac{\partial}{\partial v} H}{2\left(\sqrt{2} l^{14} + 3\sqrt{2} l^{12} z^2 + \sqrt{2} l^{10} z^4 - 5\sqrt{2} l^8 z^6 - 5\sqrt{2} l^6 z^8 + \sqrt{2} l^4 z^{10} + 3\sqrt{2} l^2 z^{12} + \sqrt{2} z^{14}\right)} \\
&- \frac{5\, l^8 z^6 \frac{\partial}{\partial z} H \frac{\partial}{\partial v} H}{2\left(\sqrt{2} l^{14} + 3\sqrt{2} l^{12} z^2 + \sqrt{2} l^{10} z^4 - 5\sqrt{2} l^8 z^6 - 5\sqrt{2} l^6 z^8 + \sqrt{2} l^4 z^{10} + 3\sqrt{2} l^2 z^{12} + \sqrt{2} z^{14}\right)} \\
&+ \frac{5\, l^6 z^8 \frac{\partial}{\partial z} H \frac{\partial}{\partial v} H}{2\left(\sqrt{2} l^{14} + 3\sqrt{2} l^{12} z^2 + \sqrt{2} l^{10} z^4 - 5\sqrt{2} l^8 z^6 - 5\sqrt{2} l^6 z^8 + \sqrt{2} l^4 z^{10} + 3\sqrt{2} l^2 z^{12} + \sqrt{2} z^{14}\right)} \\
&+ \frac{9\, l^4 z^{10} \frac{\partial}{\partial z} H \frac{\partial}{\partial v} H}{2\left(\sqrt{2} l^{14} + 3\sqrt{2} l^{12} z^2 + \sqrt{2} l^{10} z^4 - 5\sqrt{2} l^8 z^6 - 5\sqrt{2} l^6 z^8 + \sqrt{2} l^4 z^{10} + 3\sqrt{2} l^2 z^{12} + \sqrt{2} z^{14}\right)} \\
&+ \frac{5\, l^2 z^{12} \frac{\partial}{\partial z} H \frac{\partial}{\partial v} H}{2\left(\sqrt{2} l^{14} + 3\sqrt{2} l^{12} z^2 + \sqrt{2} l^{10} z^4 - 5\sqrt{2} l^8 z^6 - 5\sqrt{2} l^6 z^8 + \sqrt{2} l^4 z^{10} + 3\sqrt{2} l^2 z^{12} + \sqrt{2} z^{14}\right)} \\
&+ \frac{z^{14} \frac{\partial}{\partial z} H \frac{\partial}{\partial v} H}{2\left(\sqrt{2} l^{14} + 3\sqrt{2} l^{12} z^2 + \sqrt{2} l^{10} z^4 - 5\sqrt{2} l^8 z^6 - 5\sqrt{2} l^6 z^8 + \sqrt{2} l^4 z^{10} + 3\sqrt{2} l^2 z^{12} + \sqrt{2} z^{14}\right)} \\
&+ \frac{i\sqrt{2}\sqrt{l^2 + z^2}\, l^9 m^{\frac{3}{2}} v z^{\frac{3}{2}} \frac{\partial}{\partial v} H^2}{\sqrt{2} l^{14} + 3\sqrt{2} l^{12} z^2 + \sqrt{2} l^{10} z^4 - 5\sqrt{2} l^8 z^6 - 5\sqrt{2} l^6 z^8 + \sqrt{2} l^4 z^{10} + 3\sqrt{2} l^2 z^{12} + \sqrt{2} z^{14}} + \dots
\end{aligned}
$$

$$
\begin{aligned}
\Psi_4 = {} & \frac{l^{10} z \frac{\partial}{\partial z} H^2}{2\left(l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}\right)} + \frac{l^8 z^3 \frac{\partial}{\partial z} H^2}{2\left(l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}\right)} - \frac{l^6 z^5 \frac{\partial}{\partial z} H^2}{l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}} \\
& - \frac{l^4 z^7 \frac{\partial}{\partial z} H^2}{l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}} + \frac{l^2 z^9 \frac{\partial}{\partial z} H^2}{2\left(l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}\right)} + \frac{z^{11} \frac{\partial}{\partial z} H^2}{2\left(l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}\right)} \\
& + \frac{l^{10} z H \frac{\partial^2}{(\partial z)^2} H}{2\left(l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}\right)} + \frac{l^8 z^3 H \frac{\partial^2}{(\partial z)^2} H}{2\left(l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}\right)} \\
& - \frac{l^6 z^5 H \frac{\partial^2}{(\partial z)^2} H}{l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}} - \frac{l^4 z^7 H \frac{\partial^2}{(\partial z)^2} H}{l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}} + \frac{l^2 z^9 H \frac{\partial^2}{(\partial z)^2} H}{2\left(l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}\right)} \\
& + \frac{z^{11} H \frac{\partial^2}{(\partial z)^2} H}{2\left(l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}\right)} - \frac{2i\sqrt{2}\sqrt{l^2 + z^2} l^5 m^{\frac{3}{2}} v z^{\frac{5}{2}} H \frac{\partial^2}{\partial z \partial v} H}{l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}} - \frac{2\, l^6 m v z^3 H \frac{\partial^2}{\partial z \partial v} H}{l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}} \\
& - \frac{2\, l^4 m v z^5 H \frac{\partial^2}{\partial z \partial v} H}{l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}} + \frac{2i\sqrt{2}\sqrt{l^2 + z^2} l m^{\frac{3}{2}} v z^{\frac{13}{2}} H \frac{\partial^2}{\partial z \partial v} H}{l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}} + \frac{2\, l^2 m v z^7 H \frac{\partial^2}{\partial z \partial v} H}{l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}} \\
& + \frac{2\, m v z^9 H \frac{\partial^2}{\partial z \partial v} H}{l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}} - \frac{2i\sqrt{2}\sqrt{l^2 + z^2} l^5 m^{\frac{3}{2}} v z^{\frac{5}{2}} \frac{\partial}{\partial z} H \frac{\partial}{\partial v} H}{l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}} - \frac{2\, l^6 m v z^3 \frac{\partial}{\partial z} H \frac{\partial}{\partial v} H}{l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}} \\
& - \frac{2\, l^4 m v z^5 \frac{\partial}{\partial z} H \frac{\partial}{\partial v} H}{l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}} + \frac{2i\sqrt{2}\sqrt{l^2 + z^2} l m^{\frac{3}{2}} v z^{\frac{13}{2}} \frac{\partial}{\partial z} H \frac{\partial}{\partial v} H}{l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}} + \frac{2\, l^2 m v z^7 \frac{\partial}{\partial z} H \frac{\partial}{\partial v} H}{l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}} \\
& + \frac{2\, m v z^9 \frac{\partial}{\partial z} H \frac{\partial}{\partial v} H}{l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}} - \frac{4\, l^2 m^3 v^2 z^4 \frac{\partial}{\partial v} H^2}{l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}} + \frac{4i\sqrt{2}\sqrt{l^2 + z^2} l m^{\frac{5}{2}} v^2 z^{\frac{9}{2}} \frac{\partial}{\partial v} H^2}{l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}} \\
& + \frac{2\, l^2 m^2 v^2 z^5 \frac{\partial}{\partial v} H^2}{l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}} + \frac{2\, m^2 v^2 z^7 \frac{\partial}{\partial v} H^2}{l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}} - \frac{4\, l^2 m^3 v^2 z^4 H \frac{\partial^2}{(\partial v)^2} H}{l^{10} z + l^8 z^3 - 2\, l^6 z^5 - 2\, l^4 z^7 + l^2 z^9 + z^{11}} + \ldots
\end{aligned}
$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"